Chemical Engineering Journal 546 (2026) 180477

Contents lists available at ScienceDirect

Chemical Engineering Journal

journal homepage: www.elsevier.com/locate/cej

A quadrature-based conditional moment closure for CFD of turbulent reactive flows

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ARTICLE INFO

Keywords:
Turbulent reactive flows
Quadrature-based moment methods
Conditional moment closure
Mixing-sensitive reaction

ABSTRACT

A quadrature-based conditional moment closure (CMC) for turbulent reacting flows is formulated and coupled to a fluid dynamics solver considering both the Reynolds-Average Navier Stokes (RANS) and Large Eddy Simulation (LES) approaches to describe turbulence. The generalized quadrature method of moments (GQMOM) is used to discretize the mixture-fraction space, extending the capability of traditional quadrature-based moment methods by allowing an arbitrarily refined discretization of the mixture-fraction space. The conditional quadrature method of moments is used on top of GQMOM to obtain a reconstruction of the joint mixture-fraction–reaction-progress-variable probability distribution function. The approach was implemented into OpenFOAM leveraging the quadrature algorithms available in the OpenQBMM library. The implementation is first tested considering homogeneous and one-dimensional cases. A two-dimensional confined planar jet system is then considered, where a simple one-step, mixing-sensitive chemical reaction is included to demonstrate the coupling of the quadrature-based CMC approach with RANS and LES fluid solvers.

1. Introduction

Turbulent reacting flows play a crucial role in a variety of important industrial processes, where chemical reactions are affected by transport of the species they involve. Well-known examples are mixing-dependent reactions [1–4] and turbulent combustion [5]. In fact, the mixing of the reactants, e.g., the fuel and the oxidizer in combustion processes, must be taken into account because an intimate contact between the two at the molecular level must be realized in order for the reaction to take place effectively. For example, in combustion and other mixing-sensitive reactions, discrepancies between the mixing rate and the reaction rate can mitigate the process. To add to the complexity of the problem, in certain applications, some of the species may be fed to the system from a different phase such as a spray. As a result, also mass transfer between the two phases becomes a crucial factor.

Probability distribution function (PDF) methods [6–8] are a powerful tool to model the micromixing of the reactants and the interaction between the turbulent features of the system and the chemical reactions occurring in the system itself. More specifically, one can model the evolution of the PDF of the composition vector, i.e., the composition PDF. In many practical applications, the source terms in the PDF transport equation arise from both molecular mixing and chemical reactions [8].

The number of composition variables can be very large in a reactive system, and the direct reconstruction of their PDFs would have a notable computational cost. An efficient way to reduce the computational cost is to develop models that, instead of directly dealing with the scalar transport in the physical space, decompose it into a scalar transport relative to a conditioning variable and the transport of that conditioning variable in the physical space. The conditional moment closure (CMC) [9,10] was proposed as a method to describe non-premixed turbulent combustion, and it is based on the assumption that most of the correlations between reacting scalars can be captured by their first-order conditional moment, if an appropriate conditioning variable is considered. In CMC [9], the conditioning variable is the mixture fraction, a passive scalar that is commonly employed to describe the state of mixing of the reactants in a non-premixed combustion process. Following the CMC method, only the mixture-fraction PDF is required to extract relevant information about the reaction process. Moreover, if a β-PDF is adopted, the solution of the transport equations of the mixture-fraction moments of order 1 and 2, i.e., $\langle\xi\rangle$ and $\langle\xi^2\rangle$, is sufficient [9,11].

A limitation of this approach lies in the choice of the functional form to approximate the conditional scalar dissipation rate (CSDR): the CSDR functional form, in fact, cannot be chosen independently from

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https://doi.org/10.1016/j.cej.2026.180477
Received 4 January 2026; Received in revised form 16 July 2026; Accepted 7 August 2026
Available online 13 August 2026
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the mixture-fraction PDF in order to guarantee the conservation of the conditional mean [8,12,13].

The implementation of CMC in a computational fluid dynamics (CFD) code is a significant step toward an effective modeling of non-premixed combustion and related processes. Normally, it would be necessary to deal with the additional grid required for the mixture-fraction phase space [14], significantly impacting the overall computational cost. However, this challenge is overcome in the present work by relying on the quadrature method of moments (QMOM) to implement the CMC. The main idea behind the use of QMOM is to consider a truncated sequence of the mixture-fraction moments and then compute a Gaussian quadrature rule for the mixture-fraction PDF; the quadrature-based reconstruction of the mixture-fraction PDF is then used together with joint moments for the mixture fraction and reaction progress variable to calculate the latter [11]. This approach only requires to track a small sequence of moments of the joint mixture-fraction–composition PDF or, equivalently, of the mixture-fraction–reaction-progress-variable PDF, and no additional grid is needed for the mixture-fraction space, since such a discretization is provided by the quadrature obtained from the pure moments in the mixture fraction [15–17].

In this work, the coupling between the conditional quadrature method of moments (CQMOM) [18] and the CMC is discussed, together with the feasibility of an implementation of the method in CFD solvers. The algorithm conceptualized in [11] is described and implemented in the open-source CFD framework OpenFOAM [19], leveraging the quadrature-based moment methods we implemented in our open-source library OpenQBMM [20]. As a further improvement in the solution algorithm, the generalized quadrature method of moments [21] (GQMOM) was adopted, in order to allow a flexible discretization of the mixture-fraction space. The resulting GCQMOM-CMC algorithm is first tested in homogeneous and one-dimensional scenarios, then its usage in a CFD simulation is shown by calculating the evolution of conditional moments in a two-dimensional planar-jet reactor [22]. Two approaches are considered to couple the GCQMOM-CMC approach to a CFD solver. First the sequential solution of the flow field using a steady-state flow solver, followed by the solution of the GCQMOM-CMC model is performed to illustrate the coupling of GCQMOM-CMC to a RANS flow model [23,24]. Then, the coupled time-dependent solution of the flow field and of the GCQMOM-CMC approach is demonstrated using an LES flow solver [25–27].

2. Methodology

The theory behind turbulent mixing [8], PDF methods [7,8,28], quadrature-based moment methods [11,18,21] and CMC [9,10] is illustrated in this section.

2.1. Composition pdf, moments and conditional moments

The composition vector of the system is referred to as $\boldsymbol{\phi}$, and its ith composition variable is ϕ_i. When denoting its phase-space component as ψ_i and implying summation for repeated indices, the evolution equation of the composition PDF $f_\phi(\boldsymbol{\psi};\mathbf{x},t)$ for a reactive scalar is:

$$\frac{\partial f_\phi}{\partial t}+\frac{\partial}{\partial x_i}(\langle U_i\rangle f_\phi)+\frac{\partial}{\partial x_i}(\langle u_i|\boldsymbol{\psi}\rangle f_\phi)=-\frac{\partial}{\partial \psi_i}(\langle \Gamma_i\nabla^2\phi_i'|\boldsymbol{\psi}\rangle f_\phi)$$
$$-\frac{\partial}{\partial \psi_i}[(\Gamma_i\nabla^2\langle\phi_i\rangle) f_\phi]-\frac{\partial}{\partial \psi_i}[S_i(\boldsymbol{\psi}) f_\phi]. \quad (1)$$

The first term on the l.h.s. of Eq. (1) is the rate of change of the composition PDF; the second and third terms represent the convection in physical space due to mean (or filtered) fluid velocity $\langle U_i\rangle$ and to the scalar-conditioned velocity fluctuation $\langle u_i|\boldsymbol{\psi}\rangle$. On the r.h.s. of the same equation, the first two terms account for the transport in composition space due to molecular mixing. Those are related to the molecular diffusivity of the ith species Γ_i, to the mean composition $\langle\phi_i\rangle$ and to fluctuations around the mean ϕ_i'. The transport in composition space due to the rate of production S_i for the ith species, and therefore due to the chemical reactions, is described by the last term.

The moments of the composition PDF are obtained by multiplying the PDF $f_\phi(\boldsymbol{\psi};\mathbf{x},t)$ for a function $g(\boldsymbol{\psi})$ and integrating over all of composition space:

$$\langle g(\boldsymbol{\phi};\mathbf{x},t)\rangle=\int g(\boldsymbol{\psi})\, f_\phi(\boldsymbol{\psi};\mathbf{x},t)d\boldsymbol{\psi}. \quad (2)$$

In the following, the dependency on space and time will be implied and removed from the notation for the sake of readability. The selected form of $g(\boldsymbol{\phi})$ leads to the moments of interest. The mean composition is found with $g(\boldsymbol{\phi})=\boldsymbol{\phi}$ as expressed in Eq. (3), and the joint moments between the conditioning variable ξ and the composition vector are found by imposing $g(\boldsymbol{\phi})=\xi^k\boldsymbol{\phi}$, with k expressing the order of the moment with respect to the conditioning variable. The resulting definition is found in Eq. (4):

$$\langle\boldsymbol{\phi}\rangle=\int \boldsymbol{\psi}\, f_\phi(\boldsymbol{\psi})d\boldsymbol{\psi}, \quad (3)$$

$$\langle\xi^k\boldsymbol{\phi}\rangle=\int \xi^k\boldsymbol{\psi}\, f_\phi(\boldsymbol{\psi})d\boldsymbol{\psi}. \quad (4)$$

In this scenario, the conditioning variable is the mixture fraction $\xi\in[0,1]$, a linear combination of the elements of the composition vector. When considering binary mixing between a fuel stream and an oxidizer stream, the mixture fraction is equal to 0 in the oxidizer stream and equal to 1 in the fuel stream.

The mixture fraction is a non-reactive scalar for which a PDF $f_\xi(\zeta)$ can be defined, where ζ is its phase-space variable. The underlying hypothesis of CMC is that most of the fluctuations in the composition vector can be conditioned to the fluctuations of the mixture fraction [11]. Therefore, a conditional composition PDF $f_{\phi|\xi}(\boldsymbol{\psi}|\zeta)$ is defined, together with its moments conditioned on the mixture fraction ξ assuming a certain value ζ:

$$\langle g(\boldsymbol{\phi})|\xi=\zeta\rangle=\int g(\boldsymbol{\psi})\, f_{\phi|\xi}(\boldsymbol{\psi}|\zeta)d\boldsymbol{\psi}. \quad (5)$$

An appropriate choice of $g(\boldsymbol{\psi})$ helps to define the conditional moment of interest. If $g(\boldsymbol{\psi})=\boldsymbol{\psi}$, the first-order conditional moment vector $\mathbf{Q}(\zeta)$ required for CMC is obtained:

$$\mathbf{Q}(\zeta)=\langle\boldsymbol{\phi}|\xi=\zeta\rangle=\int \boldsymbol{\psi}\, f_{\phi|\xi}(\boldsymbol{\psi}|\zeta)d\boldsymbol{\psi}. \quad (6)$$

First-order conditional moments can be used to approximate the conditional composition PDF by resorting to the multivariate delta function $\delta(\cdot)$ as:

$$f_{\phi|\xi}(\boldsymbol{\psi}|\zeta)=\delta[\boldsymbol{\psi}-\mathbf{Q}(\zeta)]. \quad (7)$$

The combination of Eqs. (7) and (5) leads to $\langle g(\boldsymbol{\phi})|\xi=\zeta\rangle=g(\mathbf{Q})$, and if $g(\boldsymbol{\phi})=\xi^k\boldsymbol{\phi}$ the following relation is obtained:

$$\langle\xi^k\boldsymbol{\phi}\rangle=\langle\xi^k\mathbf{Q}(\xi)\rangle. \quad (8)$$

This allows the quantities of interest to be calculated given the knowledge of the mixture-fraction PDF and the first-order conditional composition vector.

2.2. Moment transport equations

The transport equation of the mixture-fraction PDF f_ξ requires a closure. Being ξ a non-reactive transported scalar, the Fokker–Planck closure [29,30] has been proven to be more reliable in capturing its evolution with respect to simpler closures such as IEM [31], therefore it is adopted in this work to model the molecular mixing of the mixture fraction. The CSDR influences the transport equation for f_ξ, and it should be recalled that a relation between its functional form and f_ξ has to exist in order to avoid errors in the unconditional scalar means [8]. First, a definition for the scalar dissipation rate is given in Eq. (9):

$$\epsilon_\xi=\frac{\langle\xi'^2\rangle}{\tau_\xi}, \quad (9)$$

where $\langle \xi'^2 \rangle = \langle \xi^2 \rangle - \langle \xi \rangle^2$ is the mixture-fraction variance and τ_ξ is a micromixing timescale related to the mechanical-to-scalar timescale ratio C_ϕ and to the local turbulent timescale [32]. When using the k-ε turbulence model in a RANS simulation, scalar fields for the turbulent kinetic energy k and its rate of dissipation ε are known, and τ_ξ can be defined as:

$$\tau_\xi = \frac{k}{2\, C_\phi\, \varepsilon}. \tag{10}$$

A different definition is needed when the algorithm is coupled with LES. In this case, τ_ξ is defined as:

$$\tau_\xi = \frac{1}{2\, C_\phi\, S}, \tag{11}$$

where S is the strain rate, related to the magnitude of the rate-of-strain tensor. If the CSDR model described by Eq. (12) is adopted,

$$\langle \varepsilon_\xi | \zeta \rangle = \varepsilon_\xi \frac{\zeta - \zeta^2}{\langle \xi \rangle - \langle \xi^2 \rangle}, \tag{12}$$

f_ξ converges to the β-PDF defined in Eq. (13), where $B(a,b)$ is the beta function,

$$f_\xi(\zeta) = \frac{\zeta^{a-1}(1-\zeta)^{b-1}}{B(a,b)}, \tag{13}$$

after a sufficiently long period of time.

The β-PDF described in Eq. (13) is characterized by the following parameters related to the mixture fraction moments $\langle \xi \rangle$ and $\langle \xi^2 \rangle$:

$$\sigma = \frac{\langle \xi'^2 \rangle}{\langle \xi \rangle - \langle \xi^2 \rangle}, \tag{14}$$

$$a = \frac{\langle \xi \rangle}{\sigma},\ b = \frac{1 - \langle \xi \rangle}{\sigma}. \tag{15}$$

It should be noted that Eq. (12) is a convenient formulation of the CSDR, and therefore of the PDF, since it keeps the f_ξ in the support $[0,1]$.

Another quantity related to f_ξ and its parameters that it is useful to define is the segregation index I_ξ. It is a canonical moment defined by:

$$I_\xi = \frac{\langle \xi^2 \rangle - \langle \xi \rangle^2}{\langle \xi \rangle\, (1 - \langle \xi \rangle)}, \tag{16}$$

and expressing the degree of segregation of the system described by the mixture-fraction PDF and its moments. The denominator of the r.h.s. of Eq. (16) is the maximum theoretical value of the mixture-fraction variance. $I_\xi = 1$ and $I_\xi = 0$ represent the limit conditions of completely unmixed fluid and perfectly mixed fluid, respectively.

Given these premises, the transport equation for the mixture-fraction moment of order k in a constant-density flow ($\nabla \cdot \mathbf{U} = 0$) is:

$$\frac{\partial \langle \xi^k \rangle}{\partial t} + \mathbf{U} \cdot \nabla \langle \xi^k \rangle - \nabla \cdot (\Gamma \nabla \langle \xi^k \rangle) = \frac{1}{\tau_\xi} \left[k \langle \xi^{k-1} \rangle \langle \xi \rangle - \langle \xi^k \rangle + \frac{k(k-1)}{2} \frac{\langle \xi^{k-1} \rangle - \langle \xi^k \rangle}{\langle \xi \rangle - \langle \xi^2 \rangle} \langle \xi'^2 \rangle \right]. \tag{17}$$

The first term on the l.h.s. is the rate of change of kth order moment $\langle \xi^k \rangle$; the second and third terms are the advection and diffusion terms, respectively. In RANS simulations based on k-ε model, the value for the diffusivity coefficient Γ is found from:

$$\Gamma = \frac{\nu_t}{\mathrm{Sc}_t} + \Gamma_L, \tag{18}$$

where Sc_t is the turbulent Schmidt number, and Γ_L is the laminar diffusivity. ν_t is the turbulent kinematic viscosity, computed as [32]:

$$\nu_t = C_\mu \frac{k^2}{\varepsilon}, \tag{19}$$

where $C_\mu = 0.09$ is a model constant. In LES simulations, Γ is found from:

$$\Gamma = \frac{\nu_{SGS}}{\mathrm{Sc}_{SGS}} + \Gamma_L, \tag{20}$$

where ν_{SGS} is the subgrid viscosity, and Sc_{SGS} is the sub-grid-scale Schmidt number. The term on the r.h.s. of Eq. (17) is the source term resulting from molecular mixing when adopting the β-CSDR [11]. An important feature of Eq. (17) is that it is closed, i.e., there is no dependence on moments of order higher than k.

The first-order conditional moment vector $\mathbf{Q}$ is modeled in CMC theory [9–11] employing the CSDR and establishing $\mathbf{Q}(t, \zeta = 0) = \mathbf{Q}_0$ and $\mathbf{Q}(t, \zeta = 1) = \mathbf{Q}_1$ as boundary conditions for the mixture-fraction phase space. For the sake of the upcoming calculations, a deviation variable $\boldsymbol{\varphi}$ is defined in such a way that its first-order conditional moment is:

$$\mathbf{q}(t, \zeta) = \langle \boldsymbol{\varphi} | \xi = \zeta \rangle, \tag{21}$$

and that the following relation is valid:

$$\boldsymbol{\phi} = \boldsymbol{\varphi} + \mathbf{Q}_0\,(1 - \xi) + \mathbf{Q}_1\,\xi. \tag{22}$$

In a reactive system, $\boldsymbol{\varphi}$ and $\mathbf{q}$ are representative of the extent of the reaction, so they will usually be set to zero at the beginning of the calculations. For the same reason, $\mathbf{q}$ will be referred as a reaction progress variable. Eqs. (21) and (22) lead to the following relations:

$$\mathbf{Q}(t, \zeta) = \mathbf{q}(t, \zeta) + \mathbf{Q}_0\,(1 - \zeta) + \mathbf{Q}_1\,\zeta \tag{23}$$

$$\langle \xi^k \boldsymbol{\phi} \rangle = \langle \xi^k \mathbf{Q} \rangle = \langle \xi^k \mathbf{q} \rangle + \mathbf{Q}_0\,\langle \xi^k (1 - \xi) \rangle + \mathbf{Q}_1\,\langle \xi^{k+1} \rangle. \tag{24}$$

It can be noticed that the phase-space boundary condition on the deviation variable is simply $\mathbf{q}(t, \zeta = 0) = \mathbf{q}(t, \zeta = 1) = 0$.

In the remainder of this work, a single-component composition vector will be considered to describe and validate the procedure, i.e., $\mathbf{Q} = Q$, $\boldsymbol{\varphi} = \varphi$, and $\mathbf{q} = q$. In general, the number of independent reactions determines the number of components of such vectors [8].

In a similar fashion to what has been done for the FP closure of the molecular mixing of the mixture fraction, the FP model for q is found, and it is related to a CSDR $\langle \varepsilon_\varphi | \varphi \rangle$ and a micromixing timescale τ_φ relative to the scalar q. This leads to the ODE describing the transport equation for the conditional moment $\langle \xi^k q \rangle$:

$$\frac{\partial \langle \xi^k q \rangle}{\partial \tau} + \mathbf{U} \cdot \nabla \langle \xi^k q \rangle - \nabla \cdot (\Gamma \nabla \langle \xi^k q \rangle) = \frac{1}{\tau_\xi} \left[k \langle \xi^{k-1} q \rangle \langle \xi \rangle - \langle \xi^k q \rangle + \frac{k(k-1)}{2} \langle \xi'^2 \rangle \frac{\langle \xi^{k-1} q \rangle - \langle \xi^k q \rangle}{\langle \xi \rangle - \langle \xi^2 \rangle} + R_q (\langle \xi^k \rangle \langle q \rangle - \langle \xi^k q \rangle) \right], \tag{25}$$

where $R_q = \tau_\xi / \tau_\varphi$ is the ratio between the micromixing timescales for ξ and q, and will be referred to as the timescale ratio. In the model presented in this work, the timescale ratio R_q is a mixing constant for the scalar q and its value is related to the supposed molecular diffusivity of the scalar. In the limit case of a non-diffusive scalar, the timescale ratio should be set to zero; in the opposite case of a high molecular diffusion, $R_q \to \infty$, causing a quick decorrelation between the deviation variable and the mixture fraction. In the following, the timescale ratio will be set to 1. This means that the characteristic times for the molecular mixing of mixture fraction and deviation variables are the same, a suitable approximation for mixing at high Reynolds number conditions [8].

Similarly to Eq. (17), the transport equation for the conditional moments is also closed. However, the reader should note that only the molecular mixing of joint moments is modeled by Eq. (25). The contribution of the reaction to the system would require an additional source term $\langle \xi^k\, S(\xi, q) \rangle$ that is unclosed in principle. The method presented in this work aims at modeling the behavior of the reaction by relying on operator splitting: after the moment transport equations are solved by the CFD code, moment inversion is performed and then the progress variable is updated, given the knowledge of the reaction mechanism [15]. Before discussing the treatment of the chemical reaction in Section 2.5, the multivariate quadrature method of moments used in this work, and the coupling between CQMOM and CMC are described in Sections 2.3 and 2.4, respectively.

2.3. Quadrature method of moments

The moment transport equations expressed in Eqs. (17) and (25) are in closed form when resorting to a β-CSDR. Therefore, an ODE solver is able to track the evolution of mixture-fraction moments and conditional moments in the system. However, a simple moment-preserving algorithm is not sufficient to capture the evolution of the system due to chemical reactions: to do so, the method also has to compute the progress variable $q(\xi)$ at every timestep and update the conditional moments accordingly.

The first step is the inversion of mixture-fraction moments to find the corresponding univariate quadrature rule. In regular QMOM, the Wheeler algorithm [33,34] allows to compute weights w_i and abscissae ξ_i of the quadrature rule that approximates an integral involving $f_\xi(\zeta)$ as a weight function. Pure mixture-fraction moments can then be calculated following Eq. (26):

$$\langle \xi^k \rangle = \sum_{i=0}^{N-1} w_i \xi_i^k, \tag{26}$$

where N is the number of nodes.

To compute the quadrature, moments of order $k = 0, 1, \dots, 2N-1$ are required, meaning that a set of $2N$ moments needs to be transported to compute a quadrature rule with N nodes. However, the approach to moment inversion has been recently extended [21] in such a way that only N moments are required, allowing an arbitrary refinement of the quadrature discretization using the same number of moments, and extending the recurrence relation used to determine the quadrature by assuming a known form of the distribution. This generalized quadrature method of moments (GQMOM) [21] has been adopted throughout all the simulations presented in this work, considering the β-GQMOM formulation [21], to refine the quadrature discretization of the mixture-fraction space without increasing the number of transported moment equations. This is particularly useful during the reaction update where nonlinear source terms are evaluated at the quadrature nodes.

2.4. Conditional quadrature method of moments

As discussed in Section 2.1, CMC describes the turbulent mixing of a reacting or non-reacting scalar conditioned on the mixture fraction, under the assumption of negligible conditional fluctuations around the conditional mean. More specifically, the conditional moments of order 1 with respect to the progress variable q are used, and they are computed from the quadrature reconstruction as:

$$\langle \xi^k q \rangle = \sum_{i=0}^{N-1} w_i \xi_i^k q_i, \tag{27}$$

where $\langle \xi^k q \rangle$ is the joint moment of order k with respect to ξ and order 1 with respect to q. Nodes (w_i, ξ_i) are obtained from the quadrature of the pure mixture-fraction moments $\langle \xi^k \rangle$ as described in Section 2.3, and q_i is the progress-variable abscissa in the ith node.

The CQMOM [18] allows the conditional abscissae q_i to be calculated when the quadrature of the pure mixture-fraction moments is known. A similar algorithm was applied to model polydisperse bubble flows [35]. Assuming to know the value of the conditional moments thanks to the resolution of their transport equations, the following linear system can be solved:

$$\begin{bmatrix} 1 & 1 & \cdots & 1 \\ \xi_0 & \xi_1 & \cdots & \xi_{N-1} \\ \vdots & \vdots & \ddots & \vdots \\ \xi_0^{N-1} & \xi_1^{N-1} & \cdots & \xi_{N-1}^{N-1} \end{bmatrix} \begin{bmatrix} w_0 & & & \\ & w_1 & & \\ & & \ddots & \\ & & & w_{N-1} \end{bmatrix} \begin{bmatrix} q_0 \\ q_1 \\ \vdots \\ q_{N-1} \end{bmatrix} = \begin{bmatrix} \langle q \rangle \\ \langle \xi q \rangle \\ \vdots \\ \langle \xi^{N-1} q \rangle \end{bmatrix}. \tag{28}$$

The solution for the set of nodes q_i with $i = 0, 1, \dots, N-1$ is unique because, by definition, mixture-fraction abscissae ξ_i are distinct, and weights w_i are positive.

In order for the calculation of the quadrature reconstruction to be successful, the truncated sequence of moments must correspond to an existing PDF on the support [0,1] [36]. If such a condition is satisfied, the set of moments is said to be realizable on [0, 1]. Although the transport of moments obeys equations in closed form, and subsequently the quadrature reconstruction is not needed to compute their evolution in time, it is still required to couple CQMOM to CMC and find conditional abscissae according to Eq. (28). Moreover, the preservation of moment realizability must be ensured also by the CFD solver used to track the moment evolution during the molecular mixing process.

Appropriate numerical schemes for the integration of source terms [37], based on a predictor–corrector Runge–Kutta scheme with adaptive time step have been proposed in the literature to ensure the robustness of QMOM methods in practical applications.

The discretization of the moment advection term may lead to unrealizable moment sequences if schemes of order higher than first are adopted [38]. For such a reason, we adopt first-order advection schemes for the moment equations in this work. However, the interested reader is directed to the relevant literature on high-order realizable schemes [39,40] which have been formulated for univariate moment problems, and require extension to multivariate problems to be applied to QMOM-CMC methods. Such an extension is left to future work.

2.5. Chemical reaction

The progress variable q can be related to any variable expressing the extent of the reaction taking place in the simulated system, such as temperature in an exothermic reaction, concentration or mass fraction of the products, etc. Consider the simple, one-step, exothermic reaction $A + B \rightarrow P$. The reaction rate equation can be easily expressed in terms of the reaction-progress variable as:

$$\frac{dq}{dt} = k(T)(1 - \xi - q)(\phi_0 \xi - q), \tag{29}$$

$$k(T) = \mathrm{Da} \exp\left[\frac{\gamma \beta q}{1 + \beta(2 - q)}\right], \tag{30}$$

where $k(T)$ is the rate constant, $\phi_0 = 1/\xi_{st} - 1$ is the mixture stoichiometry related to the stoichiometric mixture fraction ξ_{st}, Da is the Damköhler number, γ is the activation energy and β is the heat release parameter. A correction based on the reaction triangle is also imposed: conditional abscissae q_i are not allowed to become negative, nor to exceed the reactive region defined by $q = \phi_0 \xi$ and $q = 1 - \xi$, being those the theoretical upper limits for the progress variable in the case of a one-step reaction with known stoichiometry. The reaction triangle, i.e., the region of allowed (ξ, q) values determined by the given premises, is shown in Fig. 1. It should be noted that the reaction rate equation can be adapted to the system of interest when a different reaction mechanism is considered.

In the following, the complete algorithm is referred to as GCQMOM-CMC, to indicate that the Generalized Quadrature Method of Moments (GQMOM) is combined with the Conditional Quadrature Method of Moments (CQMOM) and the Conditional Moment Closure (CMC). The GQMOM procedure is used to discretize the mixture-fraction space, as discussed in Section 2.3. A visual description of the GCQMOM-CMC algorithm proposed in this work is given in Fig. 2. The algorithm is inherently transient: when acting on a frozen flow-field coming from a RANS simulation, moments are transported and reaction-progress variable is updated at every timestep but no further operations are done on the initial flow field. The algorithm performs the same operations when coupled with LES, the only difference being the flow field updated at each timestep.

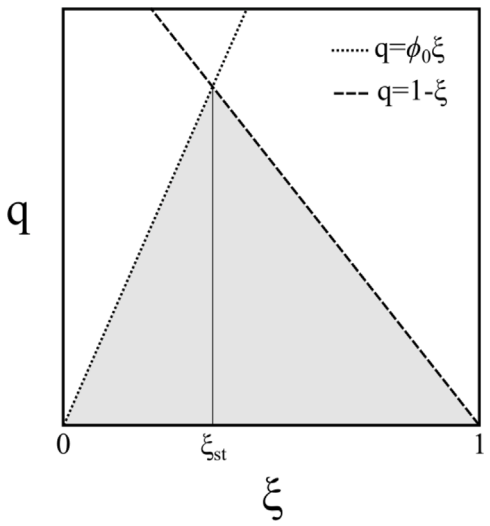


Fig. 1. Reaction triangle on the q–ξ diagram for a one-step reaction with stoichiometric mixture fraction ξ_{st} and mixture stoichiometry ϕ_0. Chemical reaction occurs only in shaded region such that q is non-negative. Complete reaction corresponds to the top of the triangle.

3. Results and discussion

The implementation of the GCQMOM-CMC approach formulated in Section 2 is demonstrated in the following sections.

In Section 3.1 a solver for a zero-dimensional problem is used to demonstrate the evolution of mixture fraction and conditional moments, together with nodes, in a homogeneous non-premixed system. The results are compared with the expected profiles obtained under analogous process conditions from a MATLAB implementation of the method already tested in [11], but with GQMOM instead of regular QMOM. Simulations are conducted to test the correct behavior of the progress variable q in two cases: in the first case, only mixing is observed while no reaction is occurring; in the second case, q represents the progress of a one-step reaction occurring together with the mixing.

In Section 3.2, one-dimensional and two-dimensional systems are studied to illustrate the behavior of the method in an actual CFD simulation.

In Section 3.3, the GCQMOM-CMC algorithm is finally coupled to a time-dependent LES flow solver. In this case, the flow field is updated at every timestep before applying the GCQMOM-CMC algorithm.

3.1. Zero-dimensional case

The moment equations for the GCQMOM-CMC approach were solved in the examples discussed in this section using the approach described in this work considering a spatially zero-dimensional domain. In this case, advection and diffusion terms are ignored, and Eqs. (17) and (25) only contain the time derivative and source term.

Univariate moments of the mixture-fraction PDF are initialized to represent a non-premixed distribution. In the following, GQMOM is always employed with $N = 10$ being the number of tracked moments, equal to the number of quadrature nodes. Appropriate initial moments are imposed at the beginning of the simulation, configuring the case of a non-premixed system with initial mean mixture fraction equal to 0.2. For this homogeneous system, the segregation index I_ξ is 0.999. The choice of non-premixed conditions results in the initial mixture-fraction moments all assuming a value close to the mean mixture fraction.

Unless otherwise specified, parameters k, C_ϕ and ε are tuned to obtain $\tau_\xi = 1$, the timescale ratio R_q is set equal to 1, the turbulent Schmidt number Sc is 0.7, the laminar diffusivity Γ_L is 10^{-6} m^2/s, and the time step is set to $\Delta t = 10^{-4}$ s.

3.1.1. Pure mixing

When q acts as a reaction-progress variable, its initial value should, in principle, be zero. However, to verify the correctness of the implemented algorithm for mixing, it is possible to initialize q to a value different from zero, and observe the transport of the progress variable due to molecular mixing. In other words, q is treated as a non-reactive scalar. A function $f_q(\xi)$ is applied to the initial mixture-fraction abscissae to obtain the initial conditions for conditional nodes, given a dependence on the mixture fraction. Subsequently, the initial conditional moments are reconstructed according to Eq. (27). In this case:

$$q_i(t=0) = f_q(\xi_i, 0) = \xi_i^6 \left(1-\xi_i\right)^2. \tag{31}$$

The evolution of moments in time is shown in Fig. 3. Only a subset of representative moments are shown in the main text for clarity; the agreement between the OpenQBMM implementation and the Matlab reference holds for the full transported moment set (see Appendix). The mixture-fraction moments start from a value close to 0.2 and evolve toward well-mixed conditions. The mean mixture fraction $\langle\xi\rangle$ stays constant throughout the entire mixing. The conditional moments increase at the beginning of the simulation accounting for the initialization of the ODE solver in Matlab and then level off. An exception is represented by the moment $\langle q\rangle$ remaining unchanged, i.e., the joint moment of order 0 with respect to ξ and order 1 with respect to q. These results are consistent with Eqs. (17) and (25): for a pure mixing problem and in absence of a reaction, the equations for $\langle\xi\rangle$ ($k = 1$ in the former equation) and $\langle q\rangle$ ($k = 0$ in the latter) reduce to $\partial\langle\xi\rangle/\partial t = 0$ and $\partial\langle q\rangle/\partial t = 0$, respectively. Therefore, the mean mixture fraction and the mean deviation variable should remain constant throughout the simulation.

Fig. 4 shows a representation of the nodes (w_i, ξ_i) and the corresponding conditional abscissae q_i. The plot on the left highlights the transition of the system to well-mixed conditions: at the beginning of the simulation, the non-premixed configuration of the system would result in the most relevant nodes in terms of weight magnitude being at the extremes of the support, 0 and 1, but as the mixing goes on, mixture-fraction abscissae progressively cluster around the mean mixture fraction. The behavior of the deviation variable during the mixing step is visible in the plot on the right: the initial distribution gradually flattens in time.

The simulation ends at 6 s, and it should be noted that the algorithm may become unstable for longer times, as the value of the mixture-fraction variance becomes too small.

3.1.2. Reaction

It is useful to recall that the GCQMOM-CMC method is based on solving the closed transport equations for mixture-fraction moments and conditional moments, performing a quadrature inversion based on GQMOM relative to the former, and exploiting the knowledge of the latter to calculate the current state of conditional nodes by solving the linear system described in Eq. (28). Before reconstructing the moments from the updated quadrature recurring to Eqs. (26) and (27), the contribution of the chemical reaction is taken into account by updating conditional abscissae following the reaction rate equation described in Eq. (29) for a single-step, exothermic reaction $A + B \rightarrow P$. In this case, the deviation variable and the conditional moments are set to zero at the beginning of the simulation, and the production rate of P is described by the dimensionless parameters in Table 1 referred to the homogeneous case.

Fig. 5 shows the behavior of a subset of representative mixture-fraction moments and conditional moments when a reaction takes place in the one-cell environment (see Supporting Information for the

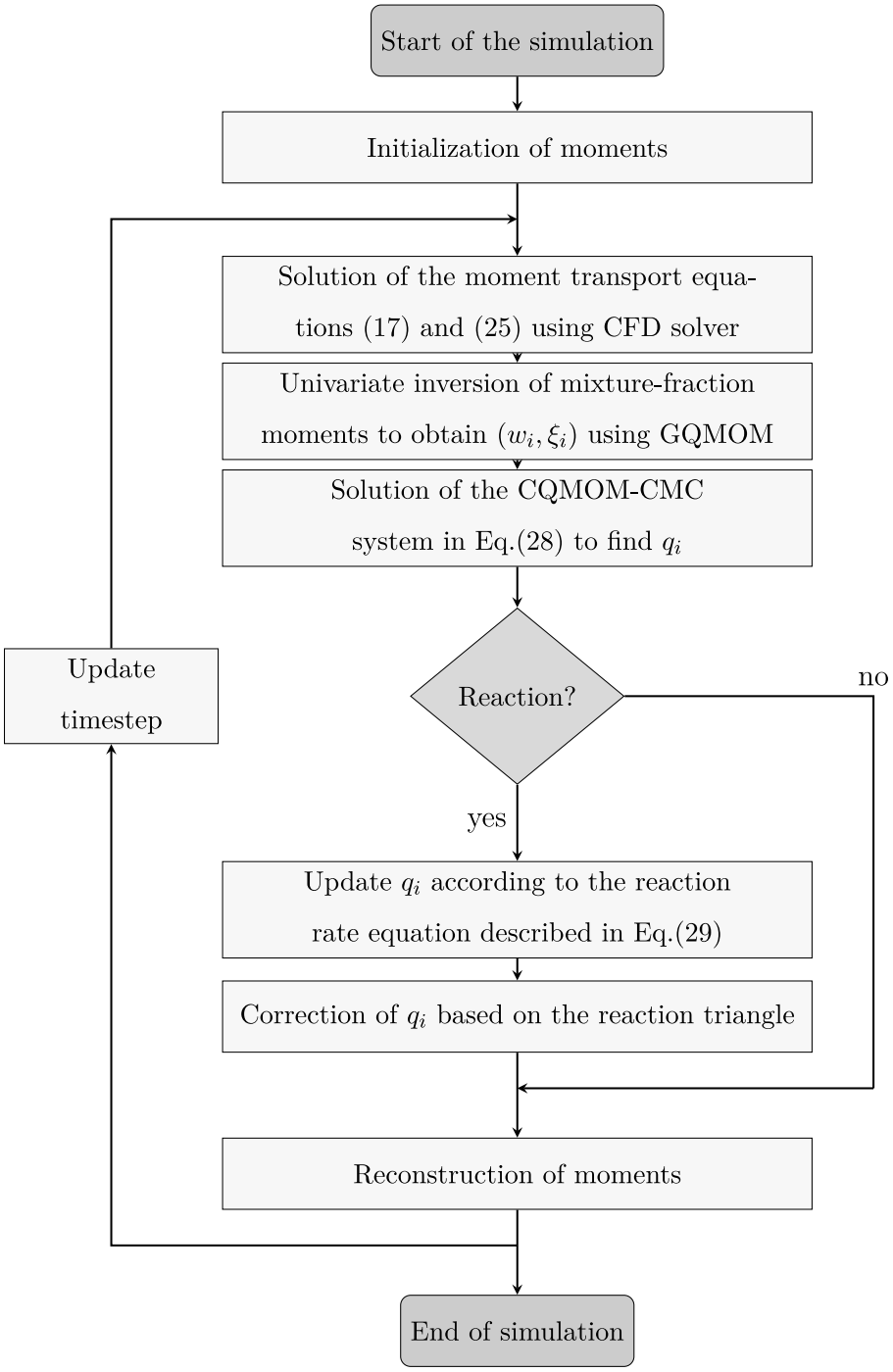


Fig. 2. Flow diagram describing the GCQMOM-CMC algorithm.

complete set of moments). The solution of the mixing problem is the same, as can be inferred from the plot on the left, while this time, together with the progress variable, conditional moments start from zero and grow according to the extent of the reaction.

Univariate and conditional nodes at selected time steps are represented in Fig. 6. The stoichiometry of the reaction defines the region in the phase space where the reaction is allowed to occur: this is taken into account in the method, limiting the growth of the reaction progress variable outside of the region.

3.2. GCQMOM-CMC with a RANS CFD flow model

This section describes how the GCQMOM-CMC approach is coupled to a flow field obtained from a RANS CFD solver. Spatially one-dimensional and two-dimensional domains are considered. Therefore, the role of advection and diffusion in the transport equations must be taken into account.

The results for three different one-dimensional test cases are reported: in Section 3.2.1, a simulation showing the impact of advection

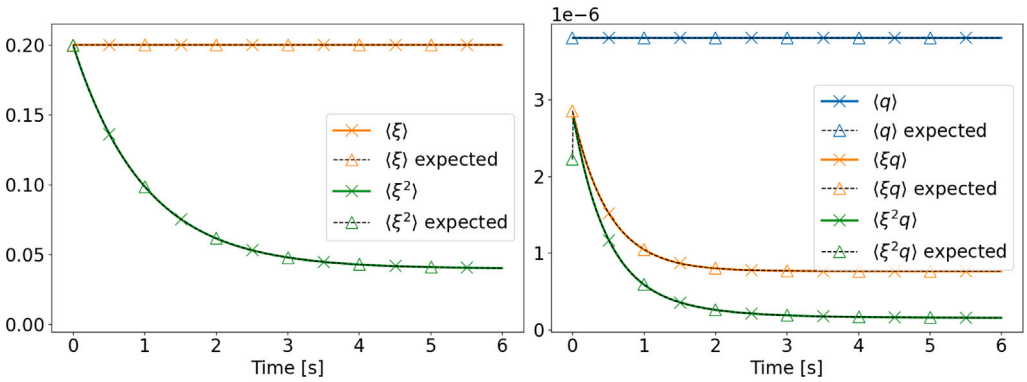


Fig. 3. Zero-dimensional case to test the mixing of q: mixture-fraction moments (left) and conditional moments (right). 0th order mixture fraction moment is not represented as it is constant and equal to 1. Colored lines: profiles from OpenQBMM simulation. Black dashed lines: expected profiles from Matlab implementation of the GCQMOM-CMC method.

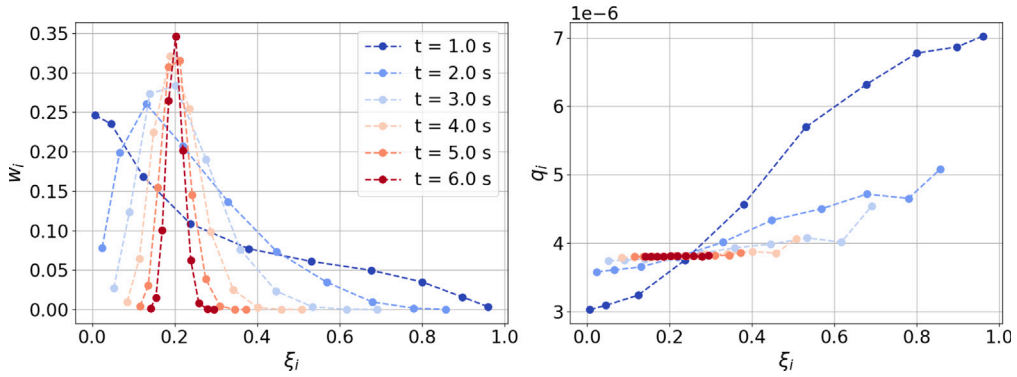


Fig. 4. Zero-dimensional case to test the mixing of q: weights w_i (left) and conditional abscissae q_i (right) with respect to mixture-fraction abscissae ξ_i at selected time steps.

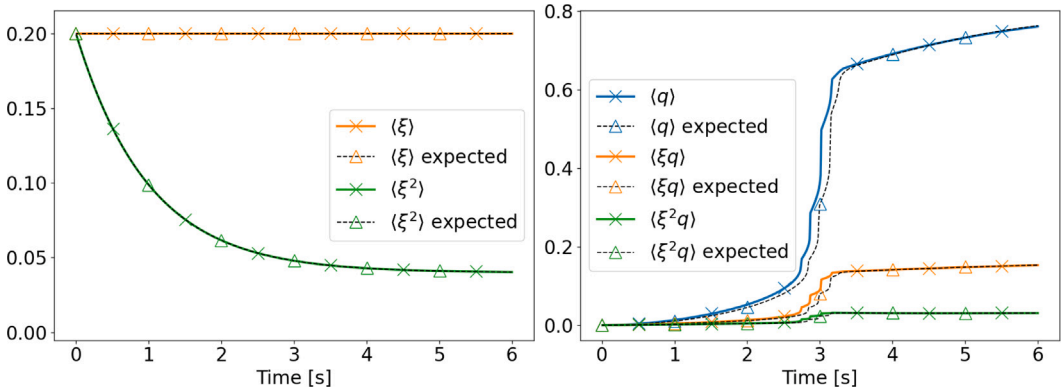


Fig. 5. Zero-dimensional case to test the behavior of a single-step exothermic reaction: mixture-fraction moments (left) and conditional moments (right). 0th order mixture fraction moment is not represented as it is constant and equal to 1. Colored lines: profiles from OpenQBMM simulation. Black dashed lines: expected profiles from Matlab implementation of the GCQMOM-CMC method.

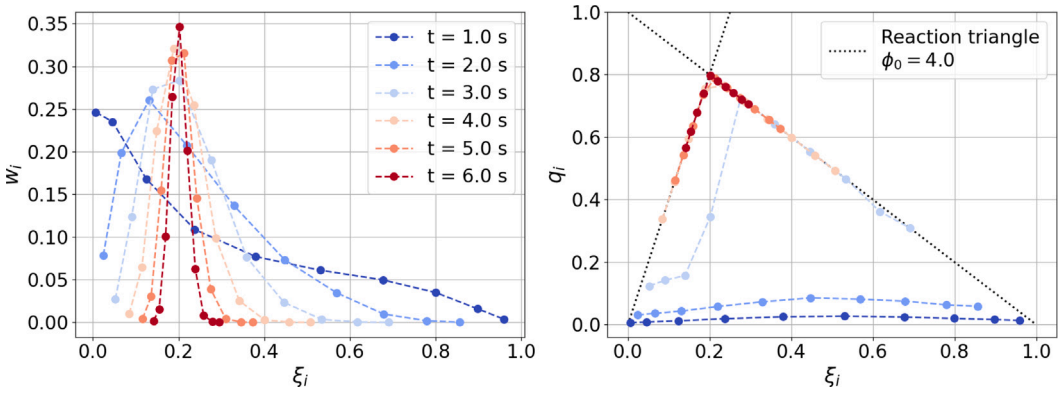


Fig. 6. Zero-dimensional case to test the behavior of a single-step exothermic reaction: weights w_i (left) and conditional abscissae q_i (right) with respect to mixture-fraction abscissae ξ_i at selected timesteps. Black dashed lines define the reaction triangle.

Table 1
Dimensionless reaction parameters describing the reaction rate for the exothermic, one-step reaction $A + B \rightarrow P$ in the homogeneous system (Section 3.1.2), in the 1D system (Section 3.2.3), and in the 2D system (Section 3.2.4).

Parameter	Symbol	Homogeneous	1D	2D
Damköhler number	Da	0.05	0.20	0.05
Activation energy	γ	30	30	10^{-5}
Heat release parameter	β	10	10	10^{-5}
Stoichiometry	ϕ_0	4	4	4

on spatially-distributed conditional moments during the mixing process; in Section 3.2.2, the evolution of a system with non-homogeneous mixing conditions along the one-dimensional domain and constant conditional abscissae; in Section 3.2.3, a reaction process occurring in a similar non-homogeneous system.

Finally, in Section 3.2.4 the solver is tested in a two-dimensional domain mirroring a reaction process in a planar jet reactor.

3.2.1. Pure mixing

Let us consider a 1D domain along x-axis between 0 and 1 m, and suppose a constant and uniform velocity field. The domain is divided into 500 cells, ensuring the Courant number to be equal to 0.05 if $U = 1$ m/s and the time step is $dt = 10^{-4}$ s. Cyclic boundary conditions are imposed at the inlet and outlet.

The mixture-fraction moments are initialized to represent a non-premixed system with mean mixture fraction $\langle\xi\rangle = 0.2$ and segregation index $I_\xi = 0.99$. The choice of $\langle\xi\rangle$ and I_ξ uniquely determines the value of the variance $\langle\xi^2\rangle$ and a Gauss–Jacobi algorithm based on recurrence coefficients for a β-PDF distribution is used to calculate a 6-node quadrature for the initialization of mixture-fraction moments. Moments are reconstructed from the quadrature following Eq. (26).

The conditional moments are initialized starting from the mixture-fraction quadrature by applying a function to mixture-fraction abscissae to calculate the corresponding initial conditional abscissae, and then applying Eq. (27). The adopted functional form is:

$$q_i = f_q(x, \xi_i) = A_q x^2 \xi_i (1 - x)^2 (1 - \xi_i) + M_1 \xi_i + M_2 (1 - \xi_i), \tag{32}$$

where x is the spatial coordinate, and A_q, M_1, M_2 are user-defined parameters. The proposed functional form f_q has the following properties: it ensures that the value of qi is greater than or equal to zero; it is dependent on space, allowing the conditional abscissae, and therefore the conditional moments, to be characterized by a spatial distribution along the one-dimensional domain. Moreover, user-defined parameters are useful for tuning the shape of the distribution, and for obtaining meaningful conditional abscissae at the extremes of the mixture-fraction phase space: in fact, the conditional abscissa q_i would approach zero when ξ_i is close to 0 or 1 because of the term $\xi_i(1 - \xi_i)$, but these are also the nodes with the highest weights in non-premixed conditions, such as the initial state of our system.

As mentioned in Section 2.4, first-order advection schemes have been used to test the method. This is done to ensure moment realizability, but causing the initial chosen distribution to get flattened during the simulation.

To study the advection of a spatial distribution of conditional moments in a mixing system, uniform mixing conditions corresponding to $I_\xi = 0.99$ are imposed along the domain. The initial conditional moments results from Eq. (32) with $A_q = 50$, $M_1 = 0$, $M_2 = 0.4$.

In this system, mixture-fraction moments evolve toward well-mixed conditions uniformly across the domain, in a manner similar to what was observed for the homogeneous system in Section 3.1.1. The spatial distribution at $t = 0$ s for the conditional moment of order zero with respect to ξ is shown in Fig. 7. The curve also shows the effect of advection on the simulation, up to a time of one second. Since the domain is 1 m long and a cyclic boundary condition is imposed at the inlet and outlet, the distribution is supposed to match the initial one exactly after this amount of time, if the velocity of the flow field is uniform and equal to 1 m/s. This occurs in our case, as can be seen by comparing the peaks of the curve at $t = 0$ s and $t = 1$ s. However, and as expected, the distribution at $t = 1$ s is noticeably flattened due to the use of first-order advection schemes.

Given that the domain is periodic and the velocity field is uniform, the difference between the profiles at $t = 0$ and $t = 1$ s directly quantifies the numerical diffusion introduced by the first-order advection scheme at the selected grid resolution.

The evolution of the quadrature nodes is shown in Fig. 8 at the probe point located in $x = 0.50$ m. Similar trends can be observed at different locations in the domain. One can notice that the distribution of w_i with respect to ξ_i at the beginning of the micromixing process reflects the bimodality of the mixture-fraction PDF for non-premixed conditions, with modes $\xi = 0$ and $\xi = 1$. As the mixing proceeds, mixture-fraction abscissae cluster around the mean mixture fraction and the variance is reduced, while conditional abscissae q_i approach a constant value.

3.2.2. Constant conditional abscissae

In the following test case, the mixing conditions are not homogeneous, but an initial sinusoidal function of the variance is imposed along the domain. Its maximum and minimum value correspond to

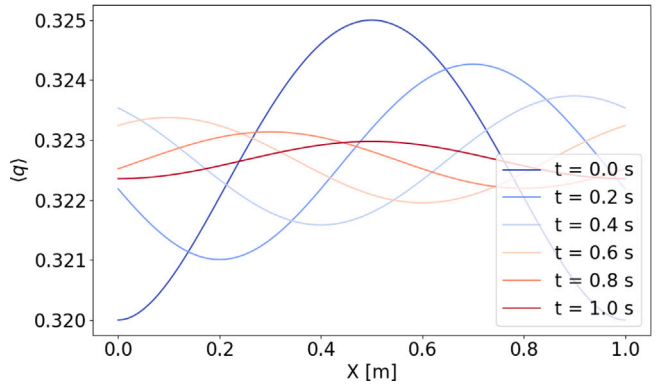


Fig. 7. Spatial distribution of the conditional moment $\langle q \rangle$ in a one-dimensional case to test advection of conditional moments during a mixing process. Color lines from blue (t = 0 s) to red (t = 1 s).

I_ξ = 0.99 at x = 0.25 m (non-premixed conditions) and I_ξ = 0.10 at x = 0.75 m (close-to-well-mixed conditions) respectively. As a result, mixture-fraction moments of order 0 and 1 are still constant and equal to 1 and 0.2 respectively, but moments of order 2 or higher have a sinusoidal profile at t = 0 s and change in time according to mixing and advection. Fig. 9 (left) shows the evolution in time of the mixture-fraction moment of order 2, i.e., the variance. The distribution is shifted along the domain because of advection and it flattens in time because mixing is occurring, but it should be kept in mind that also first-order advection schemes play a role.

A way to verify that the implemented GCQMOM-CMC algorithm maintains a correct relationship between mixture-fraction moments and conditional moments is to initialize conditional abscissae with a constant value. This can be done with an appropriate choice of the f_ξ parameters. Let us consider again Eq. (32). If A_q = 0, the dependence on the spatial coordinate x disappears, and q_i is a linear function of ξ_i from M_2 to M_1. If $M_1 = M_2$, every conditional abscissa is equal to M_1, and the kth order conditional CMC moment is related the kth order mixture-fraction moment simply by $\langle \xi^k q \rangle = M_1 \langle \xi^k \rangle$. Therefore, if $M_1 = M_2 = 1$ the kth conditional moment is supposed to be equal to the corresponding kth mixture-fraction moment along the domain and at every timestep. This is verified by comparing the left plot of Fig. 9 with the right plot showing the evolution over time of the moment $\langle \xi^2 q \rangle$. The relationship is also valid for moments of different orders. It is implied that $\langle q \rangle$ is constant, as expected when setting constant values for the quadrature abscissae representing the deviation variable.

3.2.3. Reaction

The test case that has been set to observe the evolution of the chemical reaction in a one-dimensional system is similar to the previous one described in Section 3.2.2. Again, the segregation index along the domain ranges from 0.10 to 0.99 by means of a sinusoidal distribution of the initial mixture-fraction variance $\langle \xi^2 \rangle$. The initial conditional abscissae are constant and equal to zero, i.e., A_q = 0 and $M_1 = M_2$ = 0.

The one-step, exothermic reaction $A + B \rightarrow P$ is now occurring. Therefore, q represents the extent of the reaction and $\langle q \rangle$ is no longer constant. The reaction parameters can be found in Table 1. For this scenario, the Damköhler number set to Da = 0.2, as a faster reaction better highlights the different reaction rates occurring in different regions along the domain, before these differences become less pronounced due to the general evolution of the system toward well-mixed conditions. Moreover, to limit the effect of advection, U has been set to 0.25 m/s.

The evolution in time of $\langle q \rangle$ is depicted in Fig. 10 for three probe points (indices 0, 1, 2) located at x = 0.25, x = 0.50, and x = 0.75 m respectively. Probe 2 is located in the close-to-well-mixed region where the mixture-fraction variance is $\langle \xi^2 \rangle$ = 0.056 and the progress of the reaction is faster with respect to probe 0 and 1. On the other hand, non-premixed conditions are found at probe 0 where $\langle \xi^2 \rangle$ = 0.198 at the beginning of the simulation, while probe 1 is in the middle and $\langle \xi^2 \rangle$ = 0.128. The velocity field is defined in such a way that the mixture-fraction variance is transported by advection from probe 0 to 1 in 0.5 s, and from probe 0 to 2 in 1.0 s, thus explaining the reaction extent at probe 1 being lower with respect to probe 0 after 0.5 s. It should be kept in mind that the variance distribution is not being transported without any change because of the mixing process still occurring and because of the diffusion due to first-order advection schemes. However, the reaction is fast enough to appreciate different reaction rates due to different micromixing conditions.

The influence of the micromixing status of the system on the reaction progress is also visible in Fig. 11, where the evolution of quadrature nodes in time at probe 0 and probe 2 is presented. Plots on the left shows the weights w_i with respect to the mixture-fraction abscissae ξ_i. Mixing conditions in probe 0 evolves toward well-mixed conditions because of both advection (transporting in space regions where the mixing is more pronounced, given the periodicity of the system) and mixing. On the other hand, the effect of advection and micromixing is opposite in probe 2: advection causes an initial shift toward segregated conditions, but mixing phenomena are still occurring. At t = 2.0 s, the micromixing status of the system is again close to well mixed.

The difference in the reaction rate is shown in the plots on the right, especially pointing at the magnitude of conditional abscissae at t = 0.5 s.

3.2.4. Two-dimensional planar jet

The algorithm is now tested in a 2D geometry: a confined, planar jet reactor based on the experimental and computational work by Feng et al. [22] to investigate turbulent mixing. When simplified to a 2D case, the reactor is a rectangle. The simulated fluid fed to the domain has the properties of water. The initial non-premixed conditions are given by the different characteristics of the mixture-fraction PDF associated with three separate inlets: the inner jet and two outer jets. In the inner jet, the mixture-fraction PDF is a delta function at 1, while in the outer jet it is a delta function at 0. The width of the three inlets is d = 20 mm. The flow-rate in the inner jet is 2 L/s, while for each outer jet the flow-rate is 1 L/s. This results in a Reynolds number of 50 000, i.e., highly turbulent conditions. The computational domain considered in this work is shown in Fig. 12. It is a rectangle with size 300 × 60 mm, where x is the longitudinal direction, and y is the orthogonal direction, as the three inlet jet widths combine for a total width of 60 mm. The baffles are excluded from the computational domain, and their effect is taken into account by imposing a non-uniform profile for velocity, turbulent kinetic energy and dissipation rate based on experimental results from the reference article [22]. The no-slip condition is imposed on the upper wall and on the lower wall. The computational, structured grid is made by 9800 cells, and it is refined close to walls. As in the one-dimensional case, a frozen flow field is needed to compute the mixing-dependent reaction progress using the CQMOM-CMC algorithm. In this scenario, the flow field cannot be uniform: instead, the steady-state flow field occurring in the planar jet reactor has been calculated using the solver *simpleFoam*. The k-ε model is used to address turbulence, and Eq. (19) is used to compute turbulent viscosity in every cell using values of turbulent kinetic energy and dissipation rate obtained from the model.

The resulting velocity profiles at different locations along the domain are shown in Fig. 13. Profiles are also compared with those obtained using a coarser mesh (4130 cells), a finer mesh (21 900 cells), and experimental results from Feng et al. [22]. The grid convergence is reached, and the profiles are fairly close to the expected results. It should be noted that the grid-refinement study is used here to assess the convergence of the underlying RANS flow field, for which experimental velocity data are available. The calculations involving transported

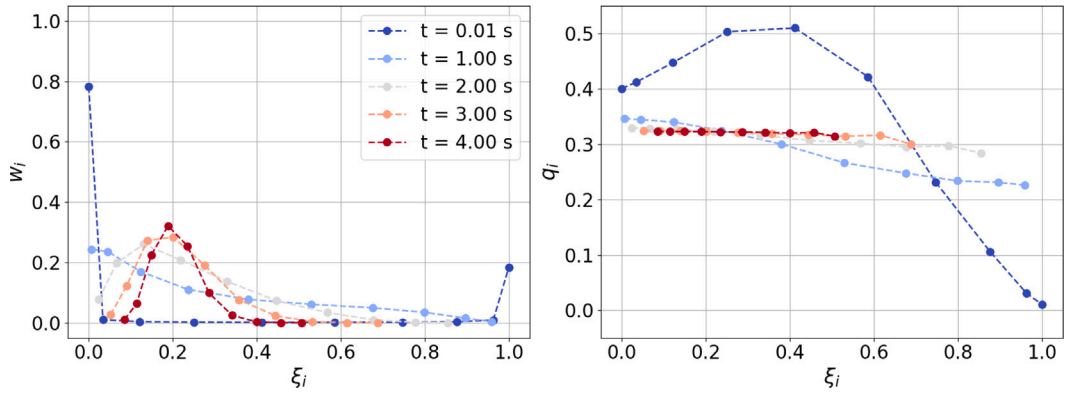


Fig. 8. one-dimensional case to test advection of conditional moments during a mixing process: weights w_i (left) and conditional abscissae q_i (right) with respect to mixture-fraction abscissae ξ_i at selected timesteps and at $x = 0.50$ m.

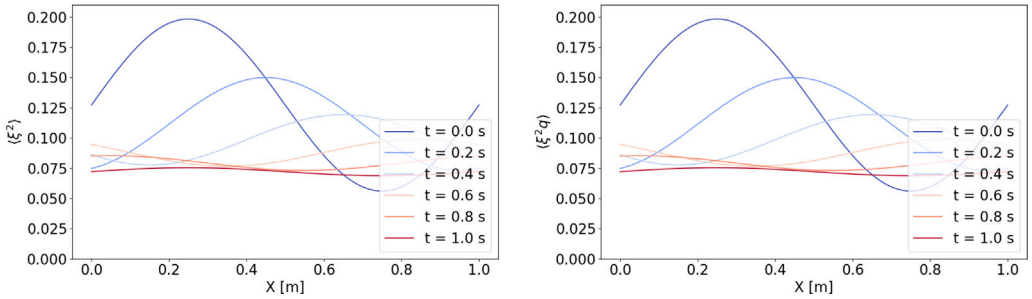


Fig. 9. Spatial distribution of the mixture-fraction moment $\langle \xi^2 \rangle$ and the conditional moment $\langle \xi^2 q \rangle$ in a one-dimensional case to test advection and mixing in non-homogeneous mixing conditions and adherence between mixture-fraction moments and conditional moments when q_i are constant. Color lines from blue (t = 0 s) to red (t = 1 s).

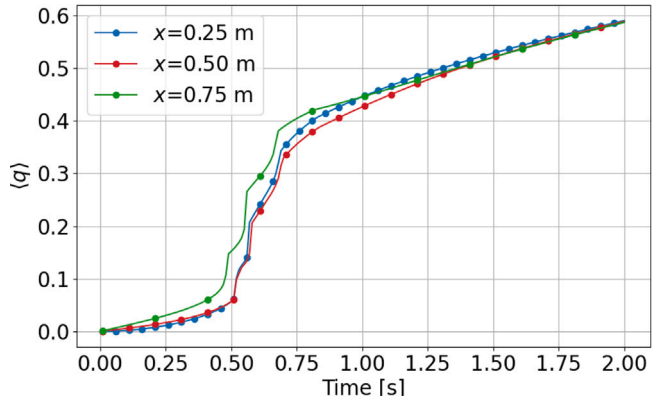


Fig. 10. Evolution in time of the conditional moment $\langle q \rangle$ at three different probe points in a one-dimensional domain with $x \in [0, 1]$ m where a chemical reaction is taking place.

mixture-fraction and conditional moments should be interpreted primarily as a demonstration of the CFD coupling and implementation feasibility of the GCQMOM-CMC algorithm.

Given the frozen flow field, the mixing transport problem is solved, together with the update of the progress variable conditioned on the mixture fraction due to the one-step reaction characterized by the following parameters, summed up in Table 1: $Da = 0.05$, $\gamma = 10^{-5}$, $\beta = 10^{-5}$, $\phi_0 = 4$. These parameters are not intended to represent a calibrated chemical system. Instead, they were selected to demonstrate the evolution of the reaction progress over the spatial and temporal scales of the planar-jet case, while limiting thermal-feedback effects and focusing on the coupling between mixing, transport, and reaction progress.

The first reactant is fed through the inner jet, while the second one is fed through the two outer jets. The mixture-fraction moment sets correspond to a PDF delta function at 1 and 0, respectively.

The resulting evolution of $\langle q \rangle$ over a time span of 0.5 s is shown as profiles along selected lines in Fig. 14. The separation of the three inlets causes the formation of two mixing layers in the domain, where the two reactants come into contact and the reaction occurs. Initially, the two mixing layers are close to each other downstream in the reactor, as apparent from the profile of $\langle q \rangle$ at $t = 0.2$ s, but are more and more defined as the system progresses toward its steady state, even far from the inlet.

The relationship between the mixing layers and the reaction progress is evident when comparing the contours of the variance $\langle \xi^2 \rangle - \langle \xi \rangle^2$ and the 0th order conditional moment $\langle q \rangle$ at the final simulated time, visible in Fig. 15.

3.3. GCQMOM-CMC with a LES CFD flow model

Large eddy simulations are intrinsically transient, which require the GCQMOM-CMC approach to be solved at each time step to ensure the correct coupling with the flow field. In this section, the PIMPLE algorithm in OpenFOAM [19], a combination of the SIMPLE [41] and PISO [42] iterative approaches, is used. The modified PIMPLE-based solver incorporating the GCQMOM-CMC approach is demonstrated by solving a LES simulation of the same planar jet analyzed in Section 3.2.4. It should be noted that, while a 2D simulation is performed here, LES are inherently 3D: therefore, the results reported here aim only to demonstrate the feasibility and robustness of the method in a simple test case. The application to more realistic 3D geometries is straightforward in OpenFOAM since the implementation of the GCQMOM-CMC is

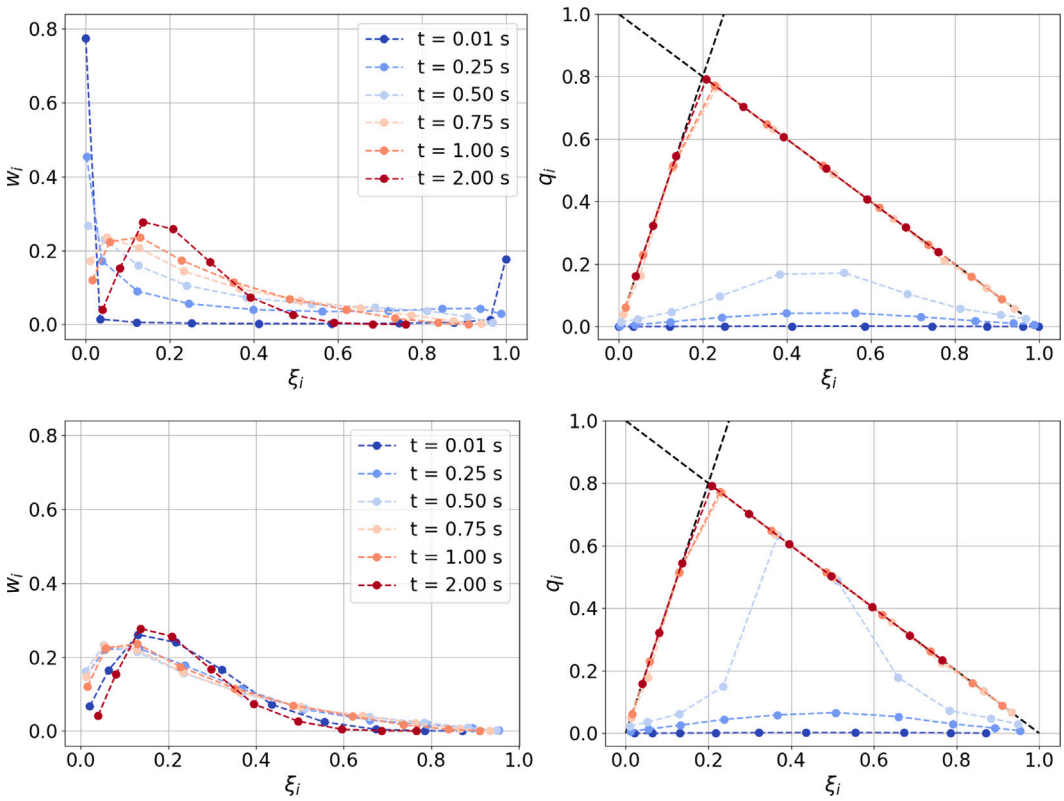


Fig. 11. One-dimensional case to test the evolution of a chemical reaction: weights w_i (left) and conditional abscissae q_i (right) with respect to mixture-fraction abscissae ξ_i at selected time steps and different probe points. Top: $x = 0.25$ m, slow reaction. Bottom: $x = 0.75$ m, fast reaction.

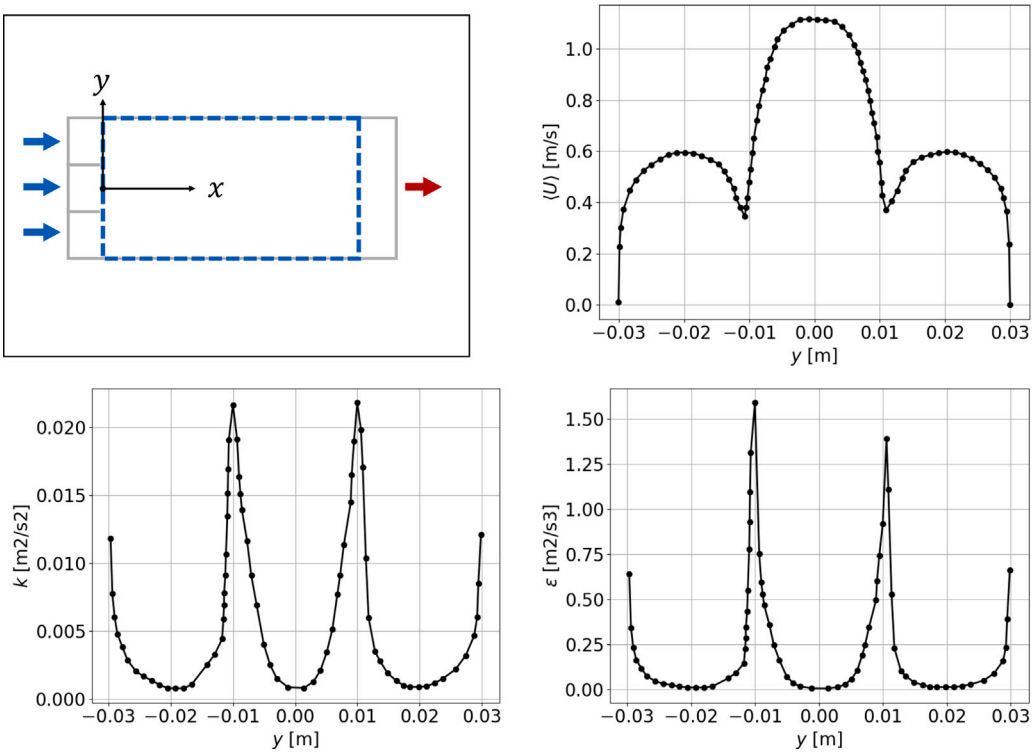


Fig. 12. 2D planar jet: schematics of the geometry (top left) and inlet conditions for velocity (top right), turbulent kinetic energy (bottom left) and its dissipation rate (bottom right).

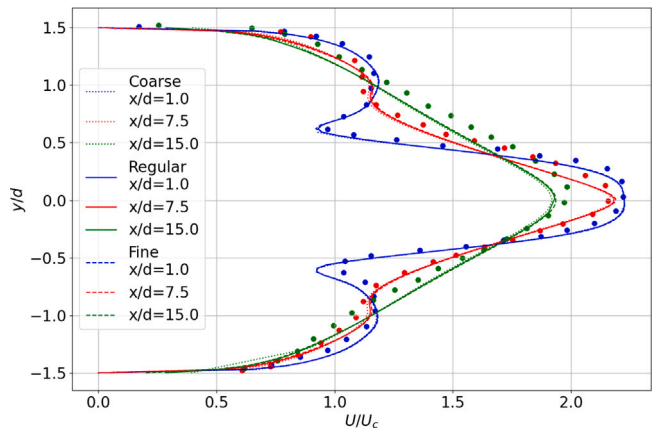


Fig. 13. 2D planar jet: velocity profiles at $x/d = 1$ (blue), $x/d = 7.5$ (red), and $x/d = 15$ (green) for three different grids: coarse (dotted lines), regular (continuous lines), fine (dashed lines) and experimental results [22].

unaffected by the number of spatial dimensions of the computational domain.

The flow field is initialized using the results obtained from the RANS simulation discussed in Section 3.2.4. The same reaction parameters are adopted. The time step is $dt = 2 \cdot 10^{-5}$ s, and the evolution of the reacting turbulent flow is simulated up to 0.3 s in this case, because of its higher computational cost. The purpose of this simulation is not to provide a direct quantitative comparison with the frozen-flow RANS simulation, but to demonstrate the fully transient coupling of the GCQMOM-CMC algorithm with the evolving LES flow field. The definition of the diffusivity coefficient Γ and the micromixing timescale τ_ξ when the algorithm is coupled with a LES simulation are reported in Eqs. (20) and (11) respectively. In these simulations, $\mathrm{Sc}_{SGS} = 0.7$.

The evolution of the field of velocity magnitude U and the field of the progress variable $\langle q \rangle$ up to $t = 0.3$ s are depicted in Fig. 16. The stream front advances along the reactor, and the progress of the reaction along the longitudinal direction and localized at the two mixing layers. The fields are also averaged considering every timestep of the simulation. The average with respect to time is represented by a bar over the variable, i.e., $\overline{U}$ and $\overline{\langle q \rangle}$. The LES is able to reproduce the progress of the reaction front in time, altogether with the computation of the unsteady flow field, thus realizing a complete coupling between the GCQMOM-CMC algorithm and LES. It should be noted that $\langle q \rangle$ can differ in magnitude from the RANS case due to the different definition of the micromixing timescale, which impacts the source term in the transport equations of momentum.

4. Conclusions

In this work, a novel algorithm conceptualized in [11] and based on the coupling between conditional moment closure and conditional quadrature method of moments has been implemented in a CFD solver based on OpenFOAM to treat reactive flows with non-premixed reactants. The computational cost of the simulations is reduced with respect to other common models: the ODE solver is only asked to deal with closed equations to transport moments, and the application of the quadrature method of moments to solve the CMC eliminates the necessity for an additional grid to deal with the mixture-fraction phase space, used in other CMC approaches. Moreover, the proposed algorithm discretizes the mixture-fraction phase space using the generalized QMOM, enhancing the refinement of the quadrature discretization at nearly the same computational effort as for QMOM.

The implementation of the GCQMOM-CMC algorithm was initially tested in a zero-dimensional problem and compared with the results obtained following the methodology described in [11] for verification purposes.

One-dimensional systems were considered to demonstrate the approach in non-homogeneous mixing conditions. A one-dimensional problem was considered to verify the implementation of moment advection and spatially dependent terms.

The method was applied to 2D simulations of planar jet reactors, considering sequential solution of the flow field with a RANS CFD solver, followed by the application of the GCQMOM-CMC approach. Finally, fully transient LES of the flow field coupled with GCQMOM-CMC were illustrated.

Further steps will be taken to improve the reliability of the algorithm and to extend the methodology to multiphase systems. The realizability of the conditional moments after the update of the progress variable is a non-trivial problem that will need to be addressed in future work. Currently, the simulations rely on first-order advection schemes to keep the transported moments realizable, but the implementation of higher-order realizable advection schemes is required to achieve reasonable accuracy in more complex simulations without compromising the realizability of the moment set [43]. Furthermore, the theoretical formulation of the method can be extended to deal with systems with pronounced multiphase features, as a disperse phase coming from a spray and transferring mass to the continuous phase, a common situation when dealing with non-premixed turbulent combustion.

CRediT authorship contribution statement

Lorenzo Vasquez Giuliano: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Alberto Passalacqua:** Writing – review & editing, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Rodney O. Fox:** Writing – review & editing, Supervision, Software, Resources, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Alberto Passalacqua reports a relationship with SimPensa LLC that includes: membership. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Complete moment verification against the Matlab implementation

In this appendix, the complete set of transported moments is shown for the two homogeneous test cases discussed in Sections 3.1.1 and 3.1.2. The purpose of these additional figures is to document the agreement between the OpenQBMM implementation and the Matlab reference implementation for all mixture-fraction moments and conditional moments considered in the simulations.

Figs. A.17 and A.18 report the full moment evolution for the pure-mixing and reacting cases, respectively. In both cases, the complete legends are included so that each curve can be identified. These results confirm that the agreement shown for the representative low-order moments is also obtained for the higher-order moments.

Data availability

No data was used for the research described in the article.

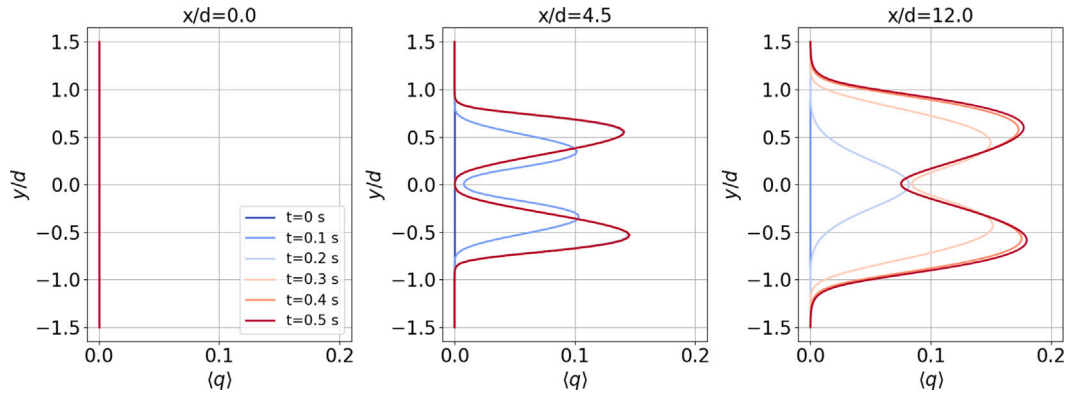


Fig. 14. GCQMOM-CMC applied on a frozen flow field in a 2D planar jet: profiles of $\langle q \rangle$ at $x/d = 1$ (left), $x/d = 7.5$ (center), and $x/d = 15$ (right) at different times.

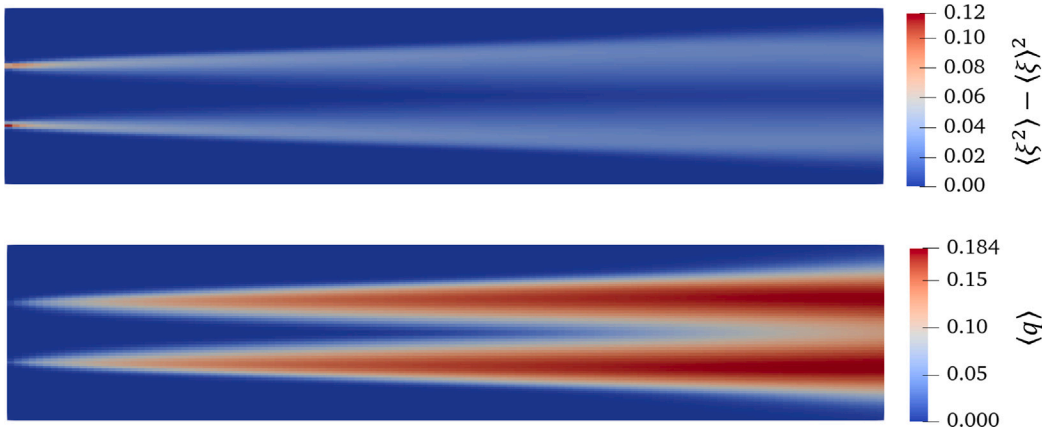


Fig. 15. GCQMOM-CMC applied on a frozen flow field in a 2D planar jet: contour of the variance $\langle \xi^2 \rangle - \langle \xi \rangle^2$ and contour of the average progress variable $\langle q \rangle$ at $t = 0.5$ s.

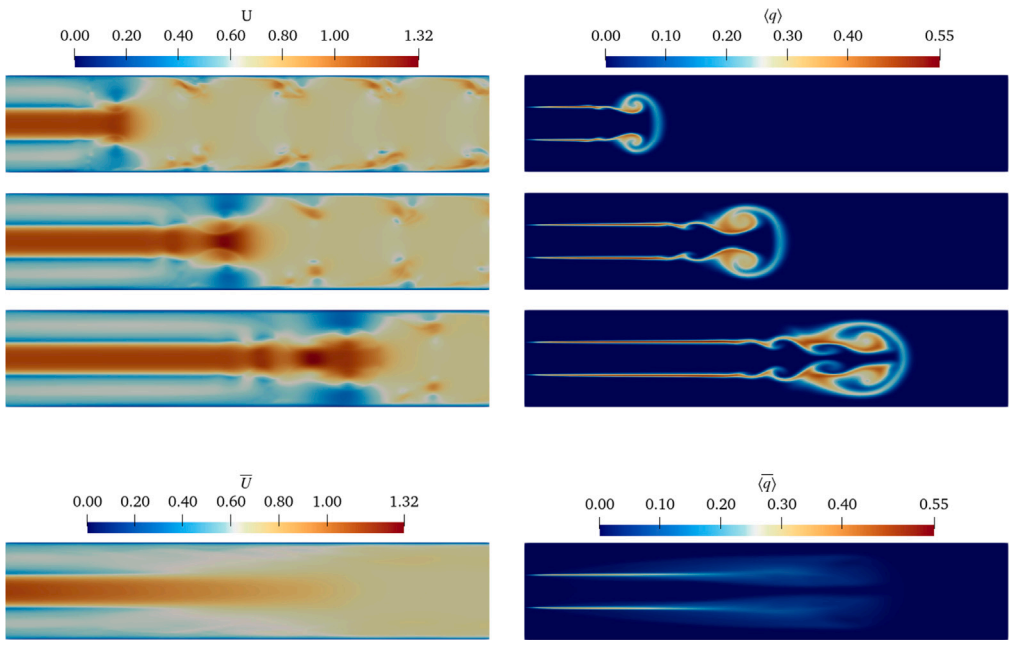


Fig. 16. LES and GCQMOM-CMC coupling a 2D planar jet: contours of U and $\langle q \rangle$ at $t = 0.1$, 0.2, and 0.3 s. The time-averaged field of velocity $\overline{U}$ and contour of the time-averaged average progress variable $\overline{\langle q \rangle}$ at $t = 0.3$ s are also represented.

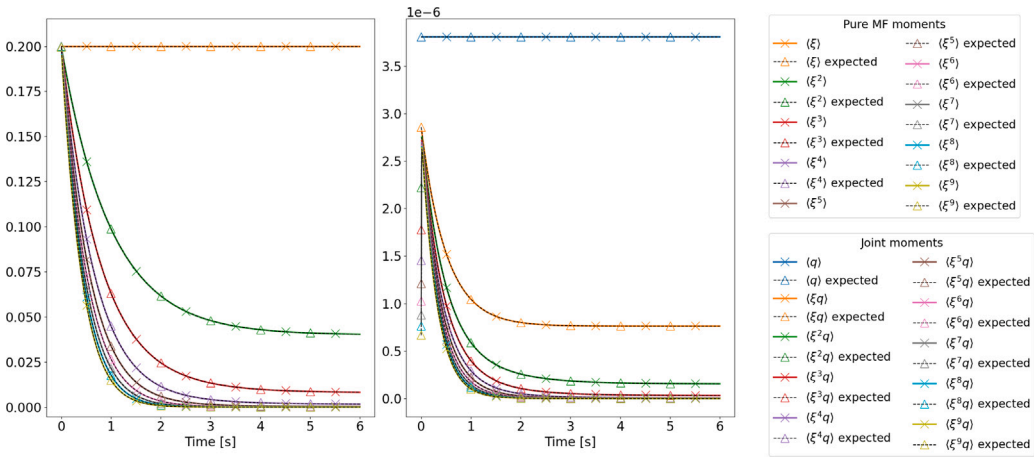


Fig. A.17. Complete verification of the homogeneous pure-mixing case. The figure reports the full set of transported mixture-fraction moments and conditional moments. The OpenQBMM results are compared with the Matlab reference implementation.

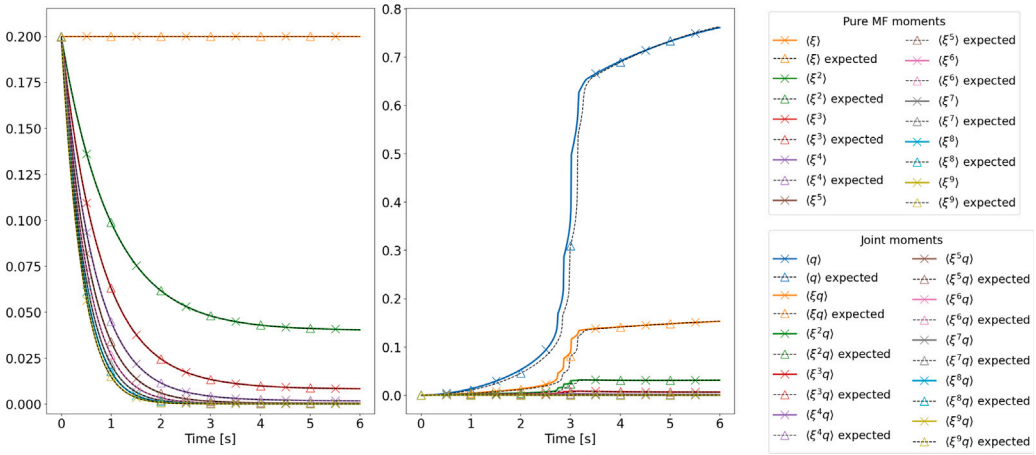


Fig. A.18. Complete verification of the homogeneous reacting case. The figure reports the full set of transported mixture-fraction moments and conditional moments for the one-step reaction test. The OpenQBMM results are compared with the Matlab reference implementation.

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